

GenAI Evidence Workbench

A Local-First Methodology for Reproducible LLM Export Forensics

PUBLIC CAPABILITY AND EVIDENCE-BOUNDARY NOTE

Version 0.1.0 | 2026-07-18

Release status

This deposit documents the public evidence contract and claim boundaries. It does not contain raw exports, conversation text, attachments, honeypot values, production detection logic, thresholds, or operational adapters.

Hiroki Tamba

Tamba Research Academy
ORCID: 0009-0004-7635-0741

Abstract

GenAI Evidence Workbench is a local-first evidence-engineering methodology for examining large exports of interactions with generative AI systems. It is designed for cases where confidentiality, provenance, reproducibility, and defensible reporting matter more than convenience. Original archives, JSON records, and attachments are treated as read-only evidence; processing boundaries and exclusions are recorded; and screening candidates are separated from findings.

This note fixes the scope and evidentiary claims of version 0.1.0. It describes the evidence contract, safe-stop policy, sanitized aggregate results from a reference case, known source limitations, and the requirements for a controlled reproduction. Because source data and production decision logic are withheld, the aggregate candidate counts cannot be independently regenerated from the public deposit alone.

1. Purpose and scope

The workbench is intended to support:

- input integrity and chain-of-custody manifests;
- streaming inspection of large ZIP and JSON exports;
- deterministic processing under a fixed implementation and configuration;
- screening for reappearance and cross-conversation lineage candidates;
- explicit handling of missing, duplicate, branched, tool, and system records;
- safe-stop exclusion when content may belong to another person; and
- sanitized reporting using locations, types, counts, and hashes instead of sensitive text.

Public boundary
The repository is a capability and evidence-boundary release. It is not an operational scanner, a public benchmark dataset, or a disclosure of production detector thresholds.

2. Reference case

An offline examination of a large ChatGPT export produced the following audited stages.

Audit stage	Count	Public interpretation
Messages in audited input set	53,203	Parsed message records under documented internal counting rules
Reappearance candidates	1,091	Screening records after normalization and exclusions
Cross-conversation lineage candidates	219	Candidate relations spanning distinct conversation identifiers

The export contained 448 conversations. One additional conversation was excluded under the safe-stop policy after indicators suggested that it might contain another person's data. The excluded conversation's content was not investigated.

The aggregate unit changes between stages: message records, screening candidate records, and candidate relations. The figures are not unique strings, confirmed incidents, or confirmed causal links.

3. Evidence contract

The methodology separates preservation from interpretation. A controlled engagement can produce:

1 Input manifest	2 Archive inventory	3 Record ledger
4 Candidate screening	5 Safe-stop / exclusions	6 Run manifest

The contract links available input bytes to enumerated archive entries, parsed records, transformations, exclusions, failures, candidate registers, and final output hashes. Each stage has an explicit counting unit.

Meaning of no unexplained data loss
This is a reconciliation claim, not an unconditional promise. It does not mean that missing source bytes were recovered or that an upstream export was complete.

4. Normalization and record boundaries

A reproducible examination must declare its handling of:

- character encoding, decoding failures, and Unicode normalization;
- timestamps, time zones, and missing time-zone information;
- exact duplicates, near-duplicates, missing fields, and malformed records;
- branched conversation graphs and user, assistant, tool, and system messages;
- attachment references and unavailable attachment bytes; and
- stable identifiers before and after normalization.

Exact production decision functions and thresholds are engagement-scoped and are not included in this release. Disclosure of the evidence contract is not disclosure of the detector.

5. Known source limitation

Truncated reference archive

The ZIP central directory was unavailable, the final local entry was partial, and 2,142,448 compressed bytes were missing. The examination reconciled recoverable complete entries and recorded the partial entry; it did not claim complete recovery.

Any record count therefore applies to the audited, recoverable input set under the declared rules, not to hypothetical bytes absent from the supplied archive.

6. Safe-stop governance

Potential third-party data triggers a safe stop. The examiner does not continue reading suspected content to improve a classification. The audit record is limited to the minimum identifier, location, type, reason code, and cryptographic hash needed to demonstrate that an exclusion occurred.

Safe-stop is intentionally asymmetric: it preserves uncertainty rather than resolving it through deeper inspection. No conclusion is drawn about the excluded content, its owner, its provenance, or the reason it appeared.

7. Candidate status and explicit non-claims

Textual similarity or metadata alone cannot establish:

- model training or training-set membership;
- model memorization, data leakage, or cross-account exposure;
- authorship, identity, or ownership;
- intentional misconduct or wrongdoing; or
- causal lineage between messages or conversations.

A metadata field such as `real_author_id` cannot authenticate identity or ownership without an independently verified account or workspace baseline. Candidate classifications require independent corroboration.

8. Public reproducibility status

This release is reproducible as a fixed public record: deposited files, the release commit, and file hashes can be independently verified. It is not a complete public reproduction package for the reference case because the source archive, sensitive records, production implementation, and thresholds are withheld.

A controlled third-party reproduction would require:

- lawful access to an authorized input export and a verified read-only copy;
- the engagement-scoped implementation, configuration, and counting specification;
- an isolated offline environment and a safe-stop escalation route; and
- comparison of run manifests and output hashes.

Interpretation

A DOI fixes the public record. It does not prove that every aggregate can be regenerated from public materials, and this release does not claim otherwise.

9. Provider-neutral evaluation roadmap

A future evaluation layer may run synthetic, sanitized, or explicitly authorized benchmark cases against multiple model providers while retaining the deterministic evidence pipeline as the system of record.

Comparisons should record provider and model identifiers, dates, parameters, prompt hashes, repetition counts, output hashes, agreement rates, abstentions, costs, and latency. Model-assisted judgments remain measurements, not ground truth.

10. Availability, license, and responsible use

Public repository:

github.com/hiroki-tamba-research/GenAI-Evidence-Workbench

The documentation and written material in version 0.1.0 are licensed under Creative Commons Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0), except where otherwise stated. The license does not grant rights to source data, customer data, private implementations, thresholds, adapters, workpapers, trademarks, or other material not included in the release.

Do not use this note to make unsupported allegations about model training, leakage, authorship, identity, or misconduct. High-stakes investigations require appropriate legal, privacy, incident-response, and subject-matter review.

References

1. Tamba, H. (2026). Non-determinism in LLM-as-Judge Graders. Zenodo. doi.org/10.5281/zenodo.20674090
2. Tamba, H. GenAI Evidence Workbench public repository. [GitHub repository](https://github.com/hiroki-tamba-research/GenAI-Evidence-Workbench)
3. UK Government. Inspect AI contribution PR 4170. github.com/UKGovernmentBEIS/inspect_ai/pull/4170

Suggested citation

Tamba, Hiroki. (2026). GenAI Evidence Workbench: A Local-First Methodology for Reproducible LLM Export Forensics (Version 0.1.0). Tamba Research Academy.

Release identifier: v0.1.0 | Language: English | Public record generated 2026-07-18